

Evaluating the Impact of Contextual Information on Human Activity Recognition

Project Report

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1 Introduction

Human Activity Recognition aims to infer a person’s activity from sensor measurements. It is a common component of wearable and Internet-of-Things systems, where the classifier must transform time-dependent signals into semantic labels such as walking or sitting. In a controlled benchmark, identifying the activity can be relatively easy when many sensors are available. A more interesting question is whether the same recognition quality can be maintained when the sensing configuration changes, for example because some body-mounted sensors are unavailable, or whether external context can provide useful complementary information.

This work follows this perspective. The project does not focus only on maximizing a single recognition score. Instead, it studies differences between sensing configurations and models. The main research question is:

Does contextual information improve activity recognition compared with an inertial-only solution, and does its contribution change when the amount of inertial information is reduced?

The study addresses four related questions:

1. What is the contribution of context-level information?

2. How much inertial information is needed for reliable HAR?
3. Does the contribution of contextual information change when fewer inertial body parts are available?
4. Can a selected model be executed on a resource-constrained embedded platform?

The experiments are intentionally comparative. A context source is considered useful not merely when it produces a high absolute score, but when it provides a reproducible gain over the corresponding inertial baseline with the same body-part configuration.

2 Dataset and sensing configurations

The project uses the RealWorld HAR dataset. The activities considered are *walking*, *running*, *sitting*, *standing*, and *lying*. The inertial sensors are an accelerometer (ACC), gyroscope (GYR), and magnetometer (MAG). They are placed on seven body parts:

Chest	Forearm	Head	Shin	Thigh	Upperarm	Waist
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The main baseline uses ACC, GYR, and MAG from all seven body parts. Contextual experiments add one or more of the following sources:

- GPS;
- light level;
- sound level from the microphone;
- the complete set of contextual sources.

The full seven-body-part baseline is deliberately important in the experimental design. It is a rich inertial representation, so it provides a strong test of whether context contains information that is genuinely complementary rather than merely compensating for missing motion data.

3 Preprocessing and feature extraction

The inertial signals are sampled at approximately 50.000 Hz. The data are segmented into windows of 150 samples, corresponding to approximately three seconds, with a stride of 75 samples and therefore 50% overlap. Each row of the final tabular dataset represents one temporal window.

For the inertial signals, time-domain statistical features include mean, standard deviation, minimum, maximum, root mean square (RMS), kurtosis, and skewness. These statistics are computed over signal components and body parts. Contextual signals are handled at window level rather than being treated as inertial signals sampled at 50.000 Hz.

The lower sampling rate of contextual sensors requires temporal alignment. In particular, GPS values are associated with inertial samples using timestamp-based backward alignment and a tolerance of approximately 30 seconds. This prevents an inertial sample from being matched with a future GPS observation. GPS information is represented using latitude, longitude, mean and maximum speed, and GPS availability within the window. The availability feature distinguishes an observed measurement from a value affected by missing data.

Missing values are handled with median imputation. The medians are estimated on the training data and then applied to the test data, avoiding information leakage. In the Python training pipeline, the preprocessing step is implemented using a `SimpleImputer(strategy=median)` followed by the Random Forest classifier.

For the embedded deployment, the current implementation focuses on the generated Random Forest inference code. The complete preprocessing pipeline, including median imputation, is not yet reproduced on the ESP32 and remains part of the future validation work.

4 Models and evaluation methodology

Three classical machine-learning models are compared: Random Forest, SVM, and XGBoost.

Random Forest was selected because it is a strong non-linear ensemble model, usually robust to heterogeneous tabular features, and because it can be converted to C for embedded deployment. SVM was included as a margin-based model that is often effective in medium-sized, high-dimensional feature spaces. XGBoost was selected as a gradient-boosted tree method capable of modelling non-linear interactions between inertial and contextual features. Together, the three models provide complementary inductive biases rather than three minor variants of the same approach.

The evaluation is organized into 15 experimental folds. In each fold, one dataset is used for training, while all remaining datasets are used for testing. The results are summarized over folds and compared at three levels:

1. global model performance;
2. activity-level behaviour and confusion patterns;
3. changes across body-part and context configurations.

The primary metric is Macro F1. For a class c , the F1-score is the harmonic mean of precision and recall:

$$F1_c = 2 \frac{\text{precision}_c \text{ recall}_c}{\text{precision}_c + \text{recall}_c}. \quad (1)$$

Macro F1 is the arithmetic mean of the class-specific F1-scores, giving every activity the same weight. This is preferable to relying only on accuracy because an activity with many samples should not dominate the evaluation. Accuracy, precision, recall, and confusion matrices are used as complementary measures.

For the context ablation, the relevant quantity is the marginal gain:

$$\Delta F1 = F1(\text{same body parts} + \text{context}) - F1(\text{same body parts, baseline}). \quad (2)$$

A positive value means that context improves the corresponding inertial configuration; a negative value means that the additional context reduces Macro F1.

5 Results

5.1 Effect of reducing the number of body parts

Figure 1 reports the mean Macro F1 as the number of inertial body parts increases. Each panel corresponds to one classifier and each curve to one context configuration. The results show that increasing the number of body parts does not necessarily lead to a monotonic improvement in performance, and that the trend depends on the classifier and context configuration.

SVM shows the clearest transition: performance rises sharply between four and five body parts, reaching approximately 0.90 Macro F1 for the baseline at five body parts. Adding a sixth or seventh body part does not improve the baseline further and slightly reduces the mean score. This indicates that, for this configuration, a compact subset can already capture most of the useful information.

Random Forest improves more gradually and remains around 0.81–0.84 Macro F1 across the progressive configurations. XGBoost also improves with additional body parts, but the increase is especially pronounced between four and five body parts. Its all-context curve reaches approximately 0.85 Macro F1 at six body parts, compared with approximately 0.75 at two body parts.

These results support the decision to analyse reduced-body-part configurations. The seven-body-part input is not automatically optimal for every model, and context should be evaluated relative to the amount of inertial information available.

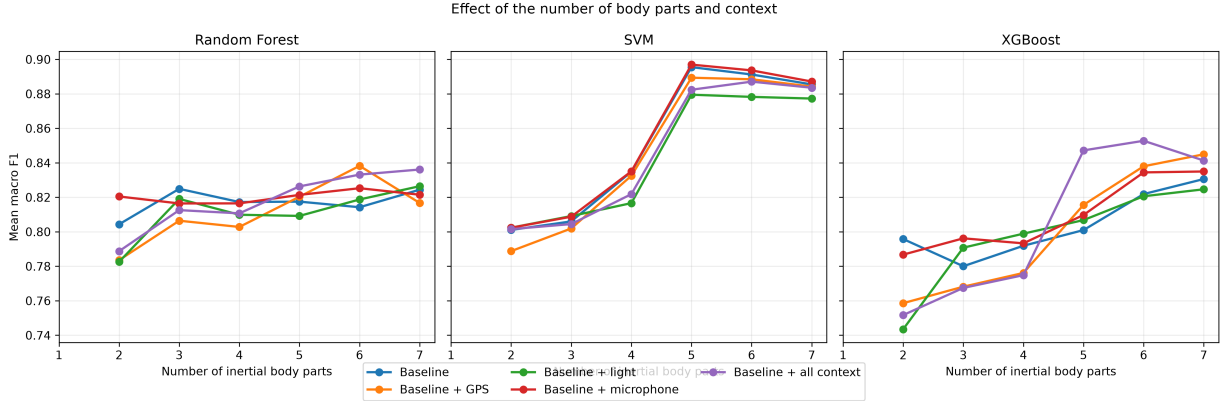


Figure 1: Mean Macro F1 as a function of the number of inertial body parts. The curves compare the baseline with GPS, light, microphone, and complete context.

5.2 Marginal value of context at equal inertial information

Figure 2 directly addresses the main experimental question. The horizontal line at $\Delta F1 = 0$ is the reference point: points above it indicate an improvement over the same body-part baseline, whereas points below it indicate a degradation. The comparison is made at equal body-part counts, so the observed difference is attributable to the contextual extension rather than to a larger inertial input.

The result is not a universal context advantage. For Random Forest, the context curves fluctuate around zero. The largest positive gains occur at six body parts, where GPS gives approximately +0.024 Macro F1 and complete context approximately +0.019. At two body parts, however, most contextual additions are negative. This suggests that the usefulness of context depends on the amount of motion information already available and is not monotonic.

SVM is comparatively insensitive to context. Most points remain close to zero, and several contextual configurations are slightly negative. This is consistent with the strong SVM baseline: once five or more body parts are available, its inertial representation is already highly discriminative. The microphone is the most promising contextual source for SVM in the complete seven-body-part comparison, but the absolute difference is small.

XGBoost exhibits the strongest context-dependent behaviour. At two body parts, context generally decreases Macro F1, but at five body parts complete context produces a gain of approximately +0.046 over the baseline. GPS and microphone also provide positive gains at five and six body parts, while light is less consistent. This suggests that XGBoost can exploit interactions between a moderately reduced inertial representation and contextual variables, whereas the same variables may be redundant or noisy at lower body-part counts.

The correct interpretation is therefore conditional: context can provide complementary information when fewer inertial body parts are available, but the effect depends on the classifier and on the specific inertial configuration. It would be incorrect to conclude that all context sources should always be added.

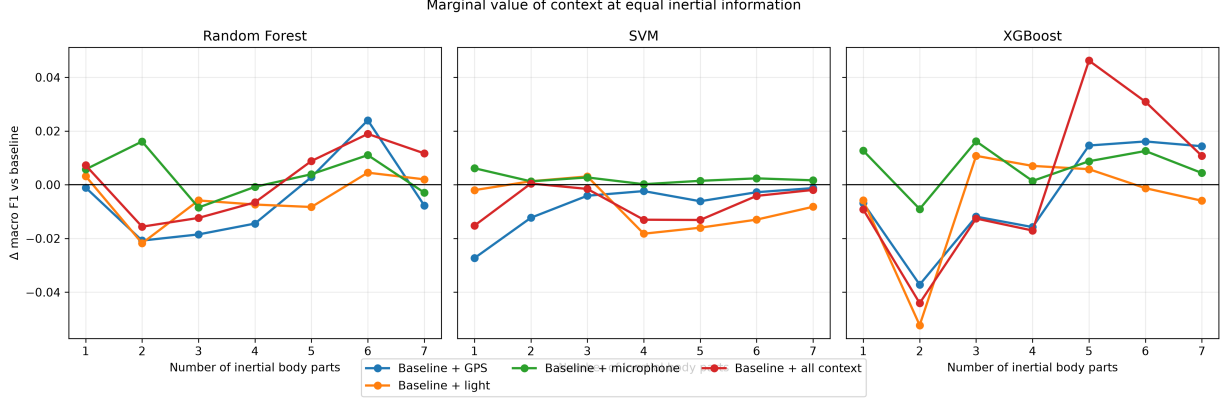


Figure 2: Marginal value of context at equal inertial information. The vertical coordinate is the difference between the contextual configuration and its body-part-matched baseline.

5.3 Informative individual body parts

Figure 3 ranks the single-body-part baseline configurations. The forearm is consistently the most informative individual body part: its Macro F1 is approximately 0.806 for Random Forest, 0.820 for SVM, and 0.813 for XGBoost. The shin is the second strongest body part for all three models.

The thigh and head are generally the weakest single-body-part configurations. For example, the thigh produces Macro F1 values of approximately 0.673, 0.674, and 0.643 for Random Forest, SVM, and XGBoost, respectively. This does not mean that these body parts are useless when combined with others; it means that their information alone is less discriminative for the five-class problem.

The ranking provides a practical insight for sensor placement. If only one inertial sensor could be retained, the forearm would be the strongest candidate among the evaluated body parts. Nevertheless, single-body-part results should not be interpreted as feature importance in a strict causal sense: a body part may be weak alone but still contribute complementary information in a multi-body-part configuration.

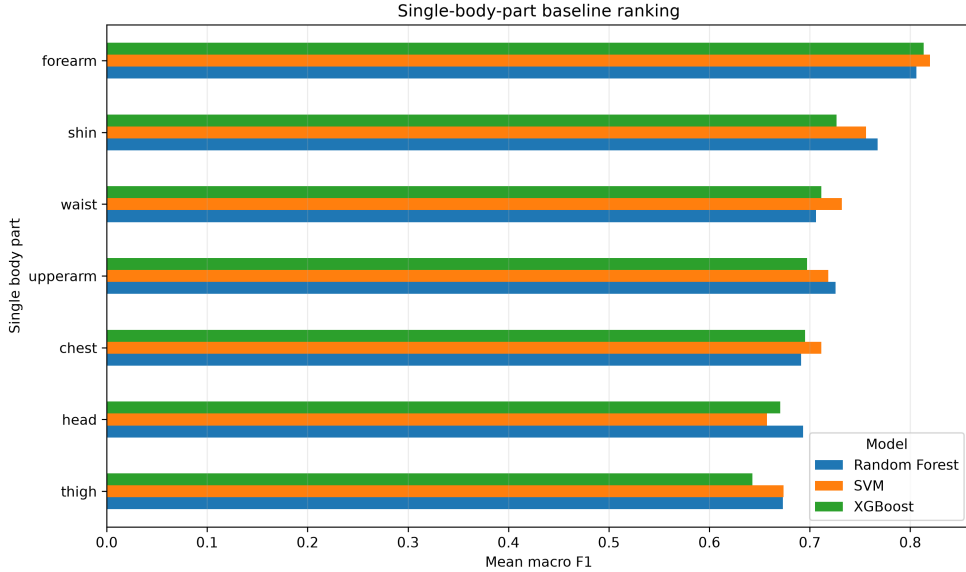


Figure 3: Macro F1 ranking for experiments using one inertial body part and no contextual extension.

5.4 Model comparison and seven-body-part reference

Figure 4 compares the three classifiers using all seven body parts and the different context configurations. SVM achieves the highest overall performance, with Macro F1 close to 0.89, while Random Forest and XGBoost remain around 0.82–0.85. This confirms that the complete seven-body-part inertial representation is already highly discriminative for the considered activities.

The addition of contextual information produces only small changes in this reference configuration. Random Forest improves from approximately 0.824 to 0.836 with complete context, while XGBoost improves from approximately 0.831 to 0.841. SVM shows an almost negligible change, from approximately 0.885 to 0.887 with the microphone. The effect of individual context sources is also model-dependent, with GPS, light, and microphone producing different gains or degradations across the classifiers.

These results provide an important reference for the reduced-body-part experiments. When all seven body parts are available, the inertial representation already captures most of the information required for the considered activities, leaving limited room for contextual information to improve performance. The larger gains observed in some reduced-body-part configurations therefore suggest that context can provide complementary information when less inertial information is available.

The three models also serve different purposes in the study. SVM provides the strongest overall performance, XGBoost shows the largest contextual gain in reduced-body-part configurations, while Random Forest offers the most practical option for embedded deployment.

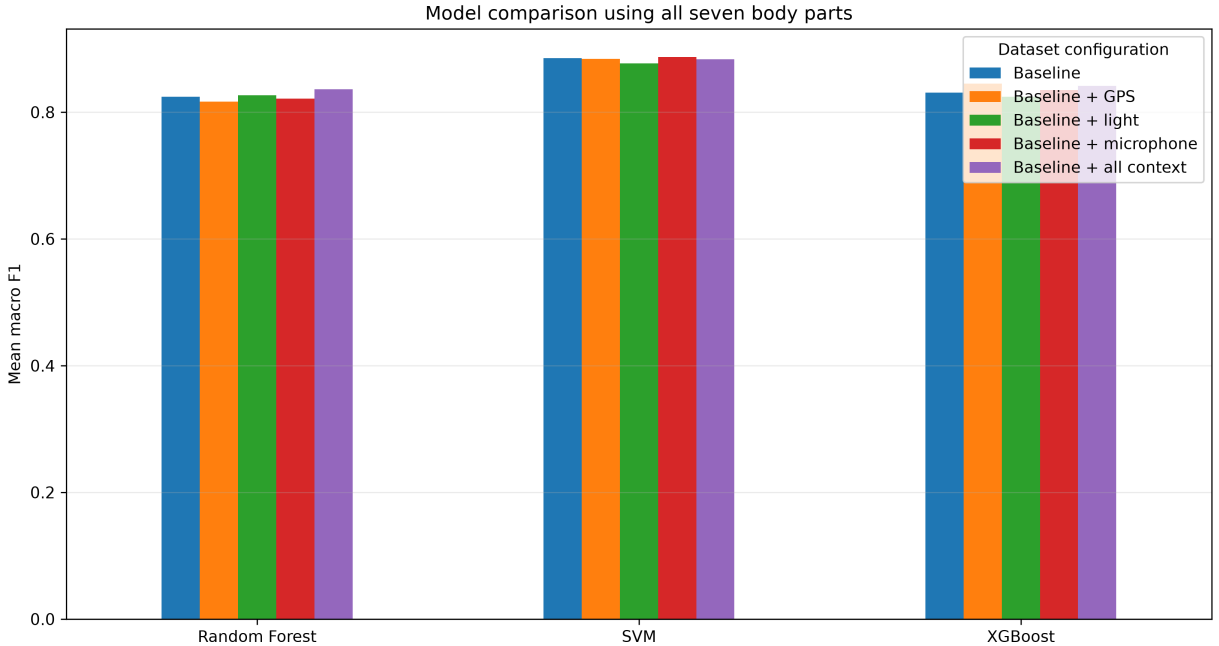


Figure 4: Model comparison using all seven body parts and the five dataset configurations.

5.5 Confusion patterns

Macro F1 summarizes the performance over all activities, but it does not show which activities are confused. To complement the aggregate comparison, Figures 5 compare the normalized confusion matrices of the baseline and complete-context configurations using all seven body parts. Rows represent the true activity and columns the predicted activity; therefore, diagonal values correspond to the recall of each activity. Values outside the diagonal indicate systematic confusions.

The matrices confirm that the contextual extension does not produce a uniform change across classes. The main differences are concentrated in the activities that are more difficult to separate, rather than in the already well-recognized classes. This is consistent with the Macro F1 results: a context source may reduce one particular confusion while having little effect on the overall

score. Consequently, the confusion matrices are used as a diagnostic complement to the aggregate metrics, not as a replacement for the fold-level comparison.

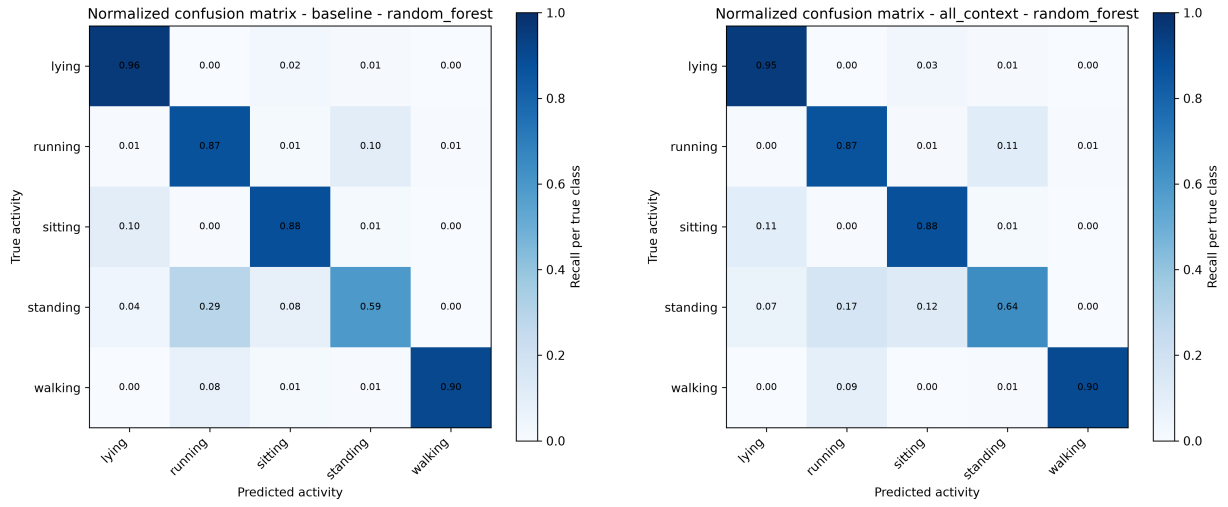


Figure 5: Random Forest normalized confusion matrices: baseline (left) and complete context (right), using all seven body parts.

The SVM matrices are reported in Figure 6, while the XGBoost matrices are reported in Figure 7.

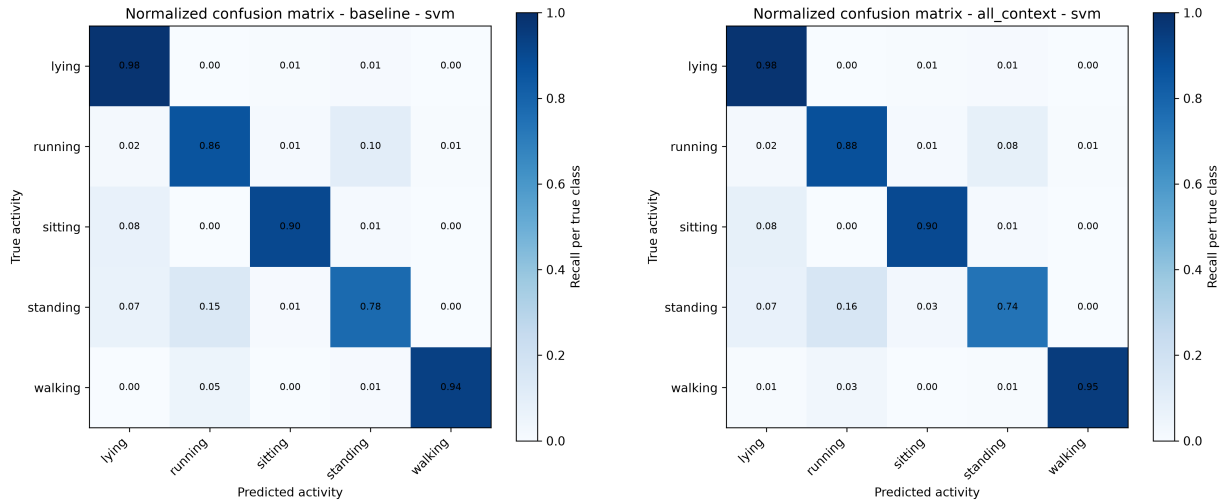


Figure 6: SVM normalized confusion matrices: baseline (left) and complete context (right).

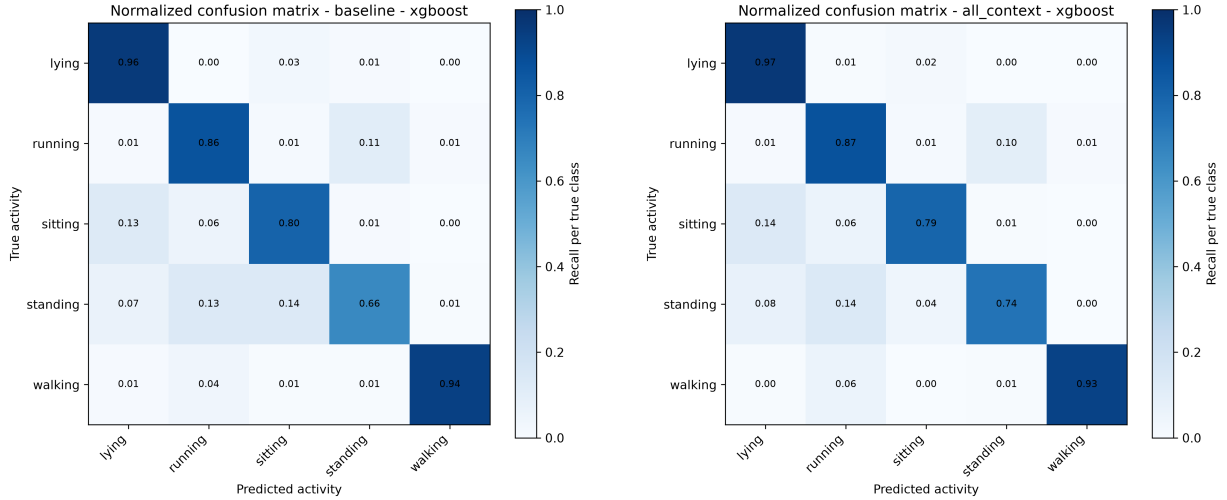


Figure 7: XGBoost normalized confusion matrices: baseline (left) and complete context (right).

6 Embedded deployment

The deployment phase focuses on Random Forest. Although SVM is the strongest model in the complete-dataset benchmark, Random Forest offers a favourable combination of non-linear modelling capacity, robustness on tabular features, and compatibility with code generation for a microcontroller. The deployment is therefore treated as a feasibility study rather than as a claim that Random Forest is the most accurate model in every configuration.

The selected Random Forest uses 455 input features, 100 trees, five output classes, unlimited maximum depth, a minimum split size of two, and a minimum leaf size of one. It was converted to C using emlearn 0.23.2 and integrated into an ESP-IDF 5.5.1 project for the ESP32-S3. The generated code contains one function for each tree and a prediction function that aggregates their outputs.

Table 1: ESP32-S3 inference benchmark.

Feature count	455
Number of trees	100
Average inference latency	1.037 ms
Minimum latency	0.931 ms
Maximum latency	2.373 ms
Throughput	approximately 964 inferences/s
Application flash	623.400 kB
DIRAM usage	53.200 kB (15.58%)
Free heap	392.800 kB

The measured average latency is approximately one millisecond, which is compatible with a window-based HAR application whose input windows span several seconds. The heap remains stable during the benchmark, suggesting that inference does not produce an observable memory leak or progressive dynamic allocation. The result demonstrates the feasibility of on-device inference for this model and feature representation.

A complete embedded evaluation should still verify prediction equivalence between the Python and ESP32 implementations. because the embedded model receives `int16_t` inputs rather than the original floating-point feature representation. Matching accuracy alone is not sufficient if feature ordering, scaling, integer conversion, or missing-value handling differ between the two environments.

7 Conclusions

This project investigated whether contextual information can improve Human Activity Recognition (HAR) and whether its contribution changes when less inertial information is available.

The results show that, with all seven body parts, inertial information is already highly discriminative and contextual sensors provide only limited and inconsistent improvements. However, when the number of body parts is reduced, context can provide complementary information in some configurations, with the strongest gain observed for XGBoost using five body parts and complete context. The effect remains dependent on both the classifier and the sensing configuration.

From an embedded perspective, the selected Random Forest was successfully deployed on an ESP32-S3, achieving an average inference latency of approximately 1.037 ms. Overall, the results suggest that contextual information is not universally beneficial, but can become valuable when combined with a reduced inertial representation.

Table 2 summarizes the reference results using all seven body parts, while Table 3 reports the largest contextual gains observed when fewer body parts are available.

Table 2: Results using all seven inertial body parts. Macro F1 values are means over the evaluation folds.

Model	Baseline	Best context	$\Delta F1$
Random Forest	0.824	0.836 (all)	+0.012
SVM	0.885	0.887 (mic)	+0.002
XGBoost	0.831	0.841 (all)	+0.011

Table 3: Best contextual gains with reduced inertial information. Macro F1 gains are computed against the baseline with the same number of body parts.

Model	Body parts	Context	$\Delta F1$
Random Forest	6	GPS	+0.024
SVM	1	Microphone	+0.006
XGBoost	5	All	+0.046

The two tables summarize the main findings of the study. With all seven body parts, contextual information provides only small gains across the three models. When fewer body parts are available, larger improvements can emerge, particularly for XGBoost. However, these gains are model and configuration dependent, showing that contextual information should be considered a complementary source rather than a universally beneficial addition.